

On certain infection problem of some graphs

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Abstract

The main topic of this article is the impact of spreading processes on graphs, which have an important role in our understanding of how information, opinions, or diseases propagate through complex networks. Graphs provide a mathematical framework to model these phenomena, where vertices represent agents and edges represent the possible interactions between them. We take into consideration a deterministic threshold-based infection process on cubic circulant graphs. The key question concerns the *contagious sets*, which are subsets of vertices whose initial infection eventually leads to the infection of all vertices in the graph. Specifically, we are interested in the infection number (or minimum contagious set size) of G :

$$m_2(G) = \min \{ |S| : S \subseteq V \text{ and every vertex in } G \text{ eventually becomes infected from } S \}.$$

This parameter provides a measure of the network's vulnerability or robustness under threshold-based diffusion. Determining $m_2(G)$ is closely related to classical problems in bootstrap percolation, threshold models of influence, and graph connectivity theory. Despite its simple definition, computing or characterizing the infection number for specific families of graphs remains a challenging combinatorial problem.

1 Problem Definition and Introduction

1.1 Cubic Circulant Graphs

The cubic circulant graphs form an important subclass of Cayley graphs on cyclic groups. They are simple, highly symmetric graphs of valency three (that is, 3-regular), and include many well-known examples such as cycles, Möbius ladders, and certain generalized Petersen graphs. A *circulant graph* is a Cayley graph on the cyclic group $\mathbb{Z}_n$. That is,

$$\text{Cay}(\mathbb{Z}_n, S)$$

has vertex set

$$V = \mathbb{Z}_n = \{0, 1, 2, \dots, n-1\},$$

and edge set

$$E = \{ \{x, x+s\} \mid x \in \mathbb{Z}_n, s \in S \},$$

where addition is taken modulo n , and $S \subseteq \mathbb{Z}_n$ is the *connection set* (or *generator set*) satisfying

$$S = -S \quad \text{and} \quad 0 \notin S.$$

A *cubic circulant graph* is a circulant graph of degree three, that is,

$$|S| = 3.$$

For even valency, the connection set S consists of pairs $\{\pm s_i\}$, yielding edges both “forward” and “backward” along the cycle. In the cubic (odd valency) case, the group must have even order n , since one of the generators must be its own inverse, namely $n/2$, the unique element of order two in $\mathbb{Z}_n$.

Hence, every cubic circulant graph has the form

$$C_n(a) = \text{Cay}(\mathbb{Z}_n, \{\pm a, n/2\}),$$

where n is even, $1 \leq a < n/2$, and $\gcd(a, n/2) = 1$.

Equivalently:

- Each vertex x is adjacent to $x + a$, $x - a$, and $x + n/2$ (all taken modulo n).
- The graph is cubic (3-regular), vertex-transitive, and connected if and only if $\gcd(a, n/2) = 1$.

Thus, the family of cubic circulant graphs coincides with the set of all Cayley graphs on $\mathbb{Z}_n$ with generating set

$$S = \{\pm a, n/2\}, \quad n \text{ even.}$$

These graphs are all connected, 3-regular, and vertex-transitive graphs on n vertices.

1.2 Infection Rules

Let $G = (V, E)$ be a simple undirected graph and let $S \subseteq V$ be the set of initially infected vertices. The infection evolves according to the following deterministic threshold rule:

- A healthy (uninfected) vertex becomes infected if at least two of its neighbors are infected.
- Once a vertex becomes infected, it remains infected permanently.

This process continues until no more vertices can be infected. The primary goal is to study which initial sets S eventually lead to all vertices being infected and to determine the minimum size of such contagious sets.

2 Graph Models and Examples

2.1 Generating Cubic Circulant Graphs

In this problem, every vertex is labeled by the n elements in the $\mathbb{Z}_n$ set. Even though this is important for generating the graphs, we can skip this limitation when looking at the infection problem by checking if two graphs are isomorphic to each other.

We have an interesting conclusion and an interesting pattern of ones and twos shows up.

For $n = 4$ the only possible value for a is $a = 1$ and we have only one type of graph, a prism. While for $n = 6$ we have 2 types of graphs, a prism and a Möbius strip. Since these 2 are the only types of graphs which are not isomorphic to each other we will work on these 2 examples only.

k	n	#graphs
2	4	1
3	6	2
4	8	1
5	10	2
6	12	1
7	14	2
8	16	1
9	18	2
$\vdots$	$\vdots$	$\vdots$

Table 1: Generator k , number of vertices n , and the number of cubic circulant graphs for small even n .

2.2 Brute-Force Approaches for Computing $m_2(G)$

For small graphs such as $n = 4$ and $n = 6$, it is easy to conclude that

$$m_2(G) = 2,$$

since at least two infected neighbors are required to infect a single vertex, and therefore the entire graph.

The simplest brute-force approach is to check all possible subsets of vertices and simulate the infection process for each. The overall worst-case complexity is

$$O(2^n \cdot n^2).$$

We can also start from the smallest possible subset size $k = 2$ and increase k until we find a subset that infects the entire graph. For each size k , there are $\binom{n}{k}$ subsets, and simulating the infection for one subset takes $O(n^2)$ time. Therefore, the total work done is

$$\sum_{k=2}^{m_2(G)} \binom{n}{k} \cdot O(n^2),$$

Since this is a really complex, we will look at a theoretical and an optimized approach to solve this problem by looking at the structure and composition of our graphs.

3 Structure of a Prism and Möbius strip

3.1 Structure of Prism Graph

Each vertex has degree 3, with exactly two neighbors in the same cycle and one neighbor connecting to the opposite cycle. The graph is vertex-transitive meaning that all vertices have the same local structure and the graph is highly symmetrical. The graph contains many 4-cycles formed by two adjacent vertices in one cycle and their corresponding vertices in the opposite cycle. Adjacent vertices in the same cycle are critical: infecting two consecutive vertices can trigger further infection.

3.2 Infection Spread in Prism Graph

In the generated Prism graphs, for larger n , we have 2 cycles, an outer and inner cycle. The smartest way to infect a whole cycle, following our rules, we need to infect every second vertex in the cycle. Infecting 2 subsequent vertices in the cycles is ineffective and it won't spread the infection. By infecting only half of the vertices, following this strategy we get the result that the smallest number to infect an odd numbered cycle is $\lceil \frac{n}{2} \rceil$. Now to spread the infection to the outer cycle, we need only 1 infected vertex and the infection cascades and infects the whole graph. The final smallest number to infect the whole graph is

$$m_2(\text{Prism}) = \left\lceil \frac{n}{4} \right\rceil + 1.$$

where we divide n by 4, since we have 2 odd numbered cycles from n vertices.

3.3 Structure of Möbius Strip Graph

RR The Möbius strip graph consists of two cycles connected by edges, but with a twist that identifies opposite vertices with reversed orientation. Unlike prism graphs, the two cycles are not independent, the twist causes local neighborhoods to overlap in a non-symmetric way. Because of the twist, infection spreading along one cycle can influence distant vertices on the other cycle earlier than in prism graphs.

3.4 Infection Spread in Möbius Strip Graph

In this type of graph, the most important part of infection should be the "twist" where the 2 cycles interlap and the infection can easily spread from the outer cycle to the inner cycle. Depending on how many "twists" the graph has, that should be the initial minimum number of infected vertices. The Möbius strip Graph has 1 twist, so the initial number of infected vertices is 1. Now we have $\frac{n}{2}$ vertices in both cycles. Since there is a direct transition from one cycle to the other in the twist, there is a common neighbour which connects these 2 cycles, we can start by infecting one of the given common neighbours. Now to infect one whole odd length cycle, as we previously stated, we need $\lceil \frac{n}{2} \rceil$ infected vertices, making every second vertex in the chosen cycle as infected. This gives us $\lceil \frac{n}{4} \rceil$ infected vertices for one chosen cycle. In this case, we do not need the 1 extra vertex as in the Prism graph, since there is a direct transition from the infected cycle to the non-infected cycle, thus cascading the infection to the whole graph. Therefore, the smallest number to infect the whole graph is:

$$m_2(\text{Möbius Strip}) = \left\lceil \frac{n}{4} \right\rceil.$$

4 Computational part

For the computational implementation of the project, we used the JupyterLab environment together with the Python programming language. The implementation relies on the libraries `NetworkX`, `NumPy`, and `Matplotlib`.

During the computational exploration of cubic circulant graphs $C_n(a)$, three distinct strategies were identified to determine the minimum contagious set size, known as the infection number $m_2(G)$.

- **Method I: Exhaustive Brute-Force Search**

This approach tests every possible subset $S \subseteq V$ of size k , starting from $k = 1$. While mathematically certain, the search space grows exponentially (2^n), making it computationally unfeasible for graphs where $n > 20$.

- **Method II: Symmetry-Based Decomposition**

This method exploits the vertex-transitivity of circulant graphs. By fixing one vertex in the initial set S and using the graph's structure (treating it as two connected cycles for prisms or a single cycle for Möbius ladders), we reduce the number of configurations to check. However, it remains a "bottom-up" approach that is often redundant.

- **Method III: Iterative Pruning via Distance Partitions**

The most advanced method, implemented in our `excentricity` analysis, takes a "top-down" approach. We start with a fully infected graph and utilize Breadth-First Search (BFS) layers to identify vertices that can be safely removed without breaking the "contagious" property of the remaining set.

4.0.1 Comparison: Pruning vs. Decomposition

The transition from Method II (Decomposition) to Method III (Pruning) represented a significant shift in efficiency. The following table summarizes why the "Excentricity" pruning method is superior:

Feature	Symmetry Decomposition (II)	BFS Pruning (III)
Logic	Building a set from scratch.	Shrinking a full infection.
Efficiency	High redundancy (isomorphic cases).	Targeted removal (linear/quadratic).
Scalability	Slows down exponentially with n .	Handles large n efficiently.
Mechanism	Static combinatorial search.	Dynamic infection influence tracking.

Table 2: Comparison of Search Strategies

The Advantage of Pruning: The pruning method focuses on the local threshold rule ($k = 2$). By using BFS layers, we create a "checkerboard" of infection. Because a vertex in a 3-regular graph only needs 2 neighbors to be reinfected, a vertex with 3 infected neighbors is "over-covered." Method III identifies these redundant vertices and removes them, ensuring the final set S is minimal while remaining contagious.

4.0.2 The Minimal Infection BFS Algorithm

The following pseudocode describes the refined process for finding the infection number using distance partitions.

Algorithm 1: Iterative Pruning via Distance Partitions

Input: Cubic circulant graph G , root vertex v_0

Output: A minimal contagious set $S \subseteq V$

Mark all vertices in G as *infected*: $S \leftarrow V$

Perform BFS from v_0 to generate distance layers $L_0, L_1, \dots, L_k$

// Step 1: Layered Pruning

for each even index $i \in \{0, 2, 4, \dots\}$ **do**

$S \leftarrow S \setminus L_i$ // Remove infection from every second layer

end

// Step 2: Iterative Refinement

Update `infection_count`(v) for all healthy $v \notin S$

Update `influence`(v) for all infected $v \in S$

Identify candidates $C = \{v \in S \mid \exists \text{ neighbor } u \text{ where } \text{infection_count}(u) = 3\}$

while C is not empty **do**

 Select $v \in C$ with the lowest influence

$S \leftarrow S \setminus \{v\}$

 Recalculate counts and update Candidate set C

end

return S

4.0.3 Conclusion

Our computational experiments demonstrate that the infection number $m_2(G)$ for cubic circulant graphs is heavily influenced by the graph's diameter and the specific jump parameters of the connection set. By moving from a brute-force search to an iterative pruning strategy based on BFS distance partitions, we reduced the search complexity from exponential to quadratic.

The patterns observed suggest that the infection number grows linearly with n . This "top-down" approach not only provides the minimum size $m_2(G)$ but also yields the explicit configuration of the contagious set, facilitating further theoretical proofs regarding the robustness of these symmetric network structures.

5 Theoretical Proof

5.1 Idea for the Prism Infection Proof

Our every Prism graph can be divided into 2 odd length cycles which are the inner and outer part of the graph. These 2 odd length cycles have $\frac{n}{2}$ vertices each. The minimum number of vertices to infect an odd length cycle of length n is $\lceil \frac{n}{2} \rceil$ if we infect every second other vertex. Now since the inner and outer cycles have $\frac{n}{2}$ vertices, for the inner or outer cycle we need $\lceil \frac{n}{4} \rceil$ infected vertices. Every vertex in the outer cycle has its own pair vertex in the inner cycle. These edges that connect the inner and outer cycles are a bridge for the infection to jump from the inner to the outer part and vice versa. Now we have either the inner or outer part of the cycle infected as a whole. We only need 1 more infected vertex in the non-infected odd length cycle to cascade this infection to the rest of the graph. This is true due to that every vertex in either cycle has its own pair of

vertex in the other cycle. Now we if we infect one vertex in the outer cycle lets say, the neighbours of this infected vertex in the outer cycle will have 1 infected neighbour in the outer cycle which we have chosen right now, and 1 infected neighbour in the inner cycle. This will cascade the infection to the entire graph. ■

5.2 Idea for the Möbius Strip Proof

If we look carefully at the structure of the graph, we can clearly conclude that there are 2 different cycle like the Prism graph, but there is a twist which connects these 2 cycles, having a direct link to transfer the infection from one to the other cycle. On top of this, every vertex in the graph also has a neighbour in the different cycle, as in the Prism graph. So the proof is similar to the Prism proof, with a different variation due to the twist. Every cycle has a $\frac{n}{2}$ vertices. Now to infect a cycle we need $\lceil \frac{n}{4} \rceil$ infected vertices, chosen at every second vertex, having a common neighbour which is not infected between them. The most important part, to infect the whole graph is to first infect one of the vertices that make up the twist. Then we can continue to infect the outer or the inner cycle, using the strategy as in the Prism infection. The crucial part here, is that we don't need the extra vertice to cascade the infection to the non-infected cycle, since the initial infection of the vertex making up the twist, cascades this infection. Thus the minimum number of infected vertices, needed to infect the whole graph is $\lceil \frac{n}{4} \rceil$. ■

6 Conclusion

Our results highlight how subtle differences in graphs can influence "infection" spread, even among graphs with the same degree and number of vertices. The pruning-based computational approach reduces the search space therefore making it a practical tool for verifying theoretical results on larger test cases.

These findings contribute to a deeper understanding of threshold-based diffusion on highly symmetric graphs and suggest directions for future work, such as extending the analysis to other classes of circulant graphs or exploring different infection thresholds. An interesting part, is how the degree of the vertices, influence the infection spread, meaning that if there are more edges between the vertices, the infection can consequently spread easier, meaning that the initial number of infected vertices may be smaller, which should be looked upon further.